Time Series Analysis

What Is a Time Series and Why Do We Analyse It?

A time series is simply a list of numbers recorded over time. The goal of time series analysis is to understand whether those numbers follow any pattern, and if so, whether we can use that pattern to predict future values.

But before we build any forecasting model, it makes sense to first ask: Is there actually something worth predicting here? How complex or unpredictable is it? That is exactly the question this code answers.

The Concept of Entropy

- Entropy, in this context, is a measure of how irregular or unpredictable a sequence of numbers is.
- If the data has low entropy, it means the numbers follow a recognisable pattern. A forecasting model has something to learn from.
- If the data has high entropy, it means the numbers behave more like random noise. No model will forecast it reliably.

The key insight: we compute entropy first because it tells us whether building a forecast model is even worth attempting.

The Two Methods Used in the Code

Approximate Entropy (ApEn)

- ApEn works by breaking the data into small overlapping windows of a fixed size, then checking: how often do similar windows appear elsewhere in the data? If the same pattern keeps showing up, the data is regular and the entropy score is low.
- Small issue: ApEn allows a window to match itself, which slightly skews the result. This is why SampEn was developed as an improvement.

Sample Entropy (SampEn)

- SampEn does the same thing as ApEn, but with one important fix: it does not allow a window to count itself as a match. This makes the result more honest and consistent, especially on smaller datasets. It is considered the more reliable of the two methods.